3.7 WQ/RHS assumptions and questions list

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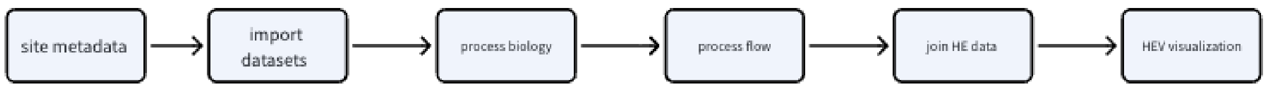
1. Current Basic Data Workflow Completed in the Dashboard

We have completed basic HE Toolkit Dashboard workflow testing in RStudio. The tested site pairing is:

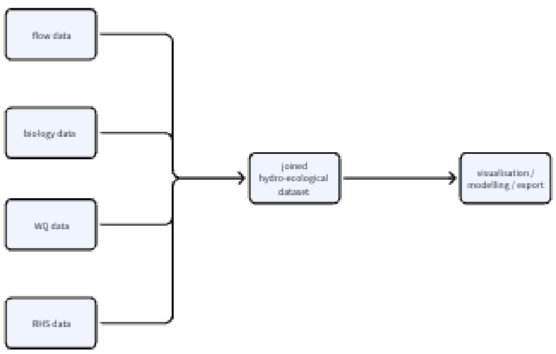
- `biol_site_id = 36183`
- `flow_site_id = F1707`
- `flow_input = HDE`

The current system workflow is:



2. Requirement Background

According to the full workflow proposed by the client, WQ should become part of the joined dataset and participate in later visualisation, modelling, and export.



3. Current Data Structure

The data currently used in RStudio mainly includes three categories.

3.1 Biology / Invertebrate Data

We have downloaded and processed Environment Agency `INV_OPEN_DATA`, and converted `INV_OPEN_DATA_METRICS.csv` into a local RDS file.

3.2 Flow Data

The current flow station is `F1707`. After import, the data is processed into flow statistics.

3.3 Environmental Data

This is mainly used for RICT predictions and includes catchment, waterbody, area, geographical, and environmental variables.

4. WQ Data Integration Assumptions

Water Quality data may include determinants such as:

- phosphate
- nitrate
- ammonia
- dissolved oxygen
- pH
- temperature
- conductivity
- suspended solids
- BOD
- other water-quality indicators

WQ data is usually time-varying because water quality changes over time, and each WQ sample normally has its own sample date. Therefore, WQ data cannot be joined only by site ID. A time-matching rule is also needed.

The initial assumption is that WQ data should be matched according to the biology sample date and then added to the joined HE dataset. In other words, each biology sample record should try to find one or more relevant WQ records.

Possible WQ matching methods include the following.

1. **Nearest Sample**

Find the WQ sample closest to the biology sample date.

Advantages:

- Simple and easy to implement.

Limitations:

- If the WQ sample is too far away from the biology sample date, it may not be representative.

2. **Seasonal Mean**

Use the mean WQ value from the same site, same year, and same season.

Advantages:

- Works well with the existing `Season` field in the current dashboard.
- Is relatively consistent with the seasonal logic used in RICT and O:E ratios.

3. **Annual Mean**

Use the annual mean WQ value from the same site and same year.

Advantages:

- More stable and less affected by single-sample outliers.

Limitations:

- May lose seasonal variation.

4. **Time-Window Summary**

Use WQ data within a time window before or around the biology sample date, such as the previous 30 days, previous 90 days, or previous 6 months.

Possible summary statistics include:

- mean
- median
- minimum
- maximum

The default window length needs to be confirmed by the user or client.

For the MVP, we could use either seasonal mean or nearest sample, and allow users to choose different matching rules in later versions.

5. RHS Data Integration Assumptions

RHS refers to River Habitat Survey data. Unlike WQ, RHS is usually not high-frequency time-series data. It is more like site-level or survey-level habitat data.

RHS may include:

- `rhs_survey_id`
- survey date
- habitat score
- channel modification score
- substrate type
- bank condition
- flow type
- vegetation structure
- habitat quality assessment variables

There may be two join methods:

1. Join directly through `rhs_survey_id`.
2. Join through site metadata mapping.

The second method is more consistent with the current dashboard because the current biology and flow relationship is also established through site metadata.

The current metadata is:

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```
biol_site_id | flow_site_id | flow_input
```

This could later be extended to:

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```
biol_site_id | flow_site_id | flow_input | wq_site_id | rhs_survey_id
```

RHS data may not be suitable for exact date matching because RHS surveys may not happen every year. A reasonable initial assumption is that RHS can first be added to the joined dataset as site-level habitat information.

If the same site has multiple RHS surveys, the selection rule needs to be confirmed. Possible rules include:

- use the RHS survey closest to the biology sample date
- use the most recent RHS survey before the biology sample date

- use the latest RHS survey
- use the RHS survey specified by the client
- summarise multiple RHS surveys

6. Assumptions for WQ/RHS Variables in Modelling

The input for the modelling MVP should be the joined dataset after the HE data join, rather than raw biology, raw flow, raw WQ, or raw RHS data.

The basic modelling MVP logic should be:

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```
joined HE dataset -> choose response variable -> choose predictor variable ->
```

After WQ/RHS are joined, they should become selectable predictor variables in the joined dataset.

Possible response variables include:

- `WHPT_ASPT_OE`
- `WHPT_NTAXA_OE`
- `LIFE_INDEX_OE`

Possible predictor variables include:

- flow metrics: `Q5` , `Q10` , `Q50` , `Q95`
- WQ variables: phosphate, nitrate, ammonia, dissolved oxygen, pH
- RHS variables: habitat score, channel modification score, substrate score

The first version of modelling should prioritise:

- numeric biology response variables
- numeric flow variables
- numeric WQ determinants
- numeric RHS scores

7. Questions to Confirm Before WQ/RHS Implementation

The following questions should be confirmed with the client or supervisor before implementing the WQ/RHS modules, to avoid building on incorrect assumptions.

7.1 WQ Questions

1. Which WQ determinants should be prioritised?

The client should confirm the most important water-quality variables. Possible candidates include phosphate, nitrate, ammonia, dissolved oxygen, pH, temperature, conductivity, and suspended solids.

2. How should WQ data be matched to the biology sample date?

The client should confirm whether to use nearest sample, seasonal mean, annual mean, or time-window summary.

3. If there are multiple WQ samples in the same time window, how should they be summarised?

The client should confirm whether to use nearest sample, seasonal mean, annual mean, or time-window summary.

4. Should there be a maximum allowed time difference?

For example, if the nearest WQ sample is more than 90 days away from the biology sample, should it still be matched, or should it be marked as missing?

5. How should WQ site IDs be mapped to biology site IDs?

The client should confirm whether WQ sites can be mapped directly through `wq_site_id`, or whether an additional site metadata mapping table is needed.

7.2 RHS Questions

1. Should RHS be joined directly through `rhs_survey_id`, or through site metadata mapping?

If the current biology site metadata does not include `rhs_survey_id`, the metadata table may need to be extended.

2. Should RHS be treated as site-level data or time-varying data?

If RHS surveys happen infrequently, RHS may be more suitable as a site-level habitat variable. If there are RHS surveys from multiple years, a time-matching rule is needed.

3. If one biology site maps to multiple RHS surveys, which one should be selected?

Possible options include selecting the nearest survey, selecting the most recent survey before the sample date, or using a survey specified by the client.

4. Which RHS variables should enter modelling?

The client should confirm which RHS indicators are numeric and ecologically meaningful, such as habitat score, channel modification score, or substrate score.

7.3 Modelling Questions

1. Which WQ/RHS variables should appear in the modelling dropdown?

For the first modelling MVP, only numeric variables should be allowed. Non-numeric variables can be used for filtering or grouping, but should not be used directly in simple regression.

2. How should WQ/RHS missing values be handled?

The system needs to decide whether incomplete rows should be removed automatically, or whether the user should first see a missing-value summary and then decide whether to continue modelling.

3. Should the dashboard show matching success rates?

Suggested matching indicators:

- number of biology samples
- number matched with WQ
- number unmatched with WQ
- number matched with RHS
- number unmatched with RHS

This would help non-coder users understand whether the joined data is reliable enough for analysis.